BBCC2026

Module 0 — From Raw Files to a Biobank-Grade Sample Sheet

- Intro — Metadata: what are we talking about?
- Part 1 — Why Metadata Matters
- Part 2 — What Good Metadata Looks Like
- Part 3 — Biobank Basics
- Part 4 — Hands-On: Build & Debug a Mini Biobank Record



- Francesco Cecere · Senior Bioinformatic Researcher · AORN Santobono-Pausilipon IRCCS
- Eva Luna Tarallo · Junior Bioinformatic Researcher · AORN Santobono-Pausilipon IRCCS

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Laboratory environment	Experimental design structure	Animal model/cell line, treatment, condition, replicates, culture condition, etc.
Environmental research	Environmental conditions	Location, temperature, sample type, sample pH, etc.
Drug development	Experimental conditions	Dose, treatment duration, administration route, test subject, etc.

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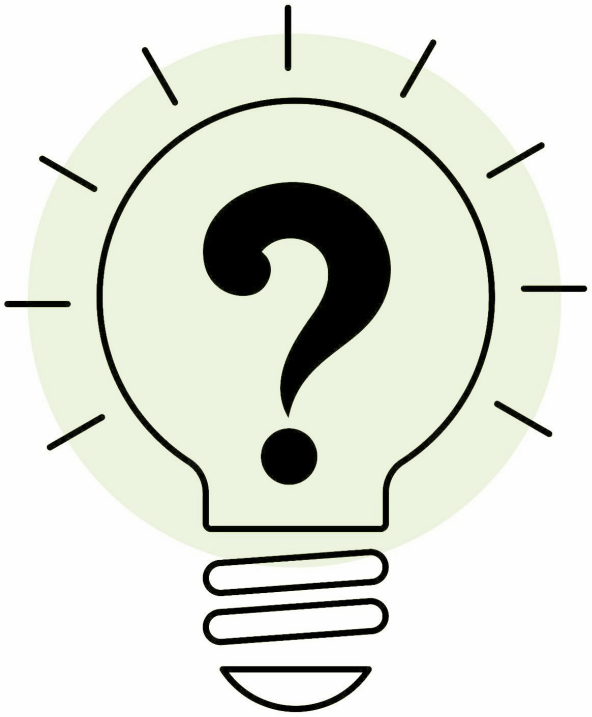
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Numbers alone are just numbers, context turn them meaningful!

This only gets more crucial in a pediatric environment!

Part 1 — Why Metadata Matters



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- PROVIDE CONTEXT
- IDENTIFY THE SAMPLE
- ENSURES CORRECT INTERPRETATION
- ENSURES DATA CONSISTENCY
- SUPPORTS REPRODUCIBILITY
- ENABLES DATA INTEGRATION



- where, when, how, and under which conditions the data were generated
- Unique identifiers are key in biobanking and sample storage
- Same numbers, different meaning upon different age, cell type, tissue, etc.
- Less data cleaning required, easier computational analysis
- Same protocol applicable
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But they’re man-made, so...



Metadata errors are often hidden!



Garbage in, garbage out!



EX. 1

Six-patient toy cohort

Part 2 — What Good Metadata Looks Like

Field	Why it is non-negotiable
Patient/subject ID	The single key linking a sample to its clinical context — must be unique, always.
Sample type	Tumor tissue, blood, saliva, etc. — determines what analyses are even valid
Collection date	Anchors the sample in time relative to diagnosis, treatment, and the patient's age.
Diagnosis	Ideally from a controlled vocabulary, not free text.
Age at collection	Needed for almost any biological interpretation.
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Any critical points?

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Platform aimed at managing clinical data implement strategies to avoid this variability!

Age is a calculated field, diagnosis can only be selected between Ontology terms.

As a general rule, any text field that can be turned into a multiple choice, is one less source of error.

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This can be done in any spreadsheet!

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- **FAIR principles** — data and metadata should be **Findable, Accessible, Interoperable, and Reusable**. In practice, for this tutorial, that mostly means: consistent IDs, documented units, and machine-readable fields rather than prose.

- **Ontologies**, such as the Human Phenotype Ontology (HPO) for clinical phenotypes, give every term a stable, shared identifier (e.g. an HPO code) instead of free text. "Kid kidney tumor" and "pediatric renal neoplasm" are the same thing to a clinician but look like two different diagnoses to a computer — an ontology term fixes that.

Part 2 — What Good Metadata Looks Like

Messy: mixed fields, inconsistent formatting, ambiguous blanks.

Patient ID	Age_sex	Notes
s1	4,girl	wilms
s2	unknown	
s3	6 yo M	neuroblastoma??

Clean: one column per variable, explicit NA, standardized terms.

patient_id	age_years	sex	diagnosis
PT-9001	4	F	Wilms tumor
PT-9002	NA	NA	NA
PT-9003	6	M	Neuroblastoma

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There's one thing that does cause computational error,
can you spot it?



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The field name has space separated words!

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If you're working with **pre-existing data**, this means you need more cleaning steps before starting the analysis.



If you're generating your own data, a multidisciplinary approach is essential to ensure that metadata can include all vital information, but also can be correctly structured for computational purpose.

Part 2 — What Good Metadata Looks Like

We’ve talked about the structure of good metadata.

But **how do we decide which metadata fields to include?**



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There's no universal metadata set that works for every occasion.

Experimental design and study goal determine what information are vital for your specific case.

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We've talked about the structure of good metadata.

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If someone else gets access to these data in five or ten years, would they have all the necessary info to analyse and contextualize it?



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The multidisciplinary approach ensures metadata are both **well formatted** and **complete**.

There's no universal metadata set that works for every occasion.

Experimental design and study goal determine what information are vital for your specific case.

Part 2 — What Good Metadata Looks Like



How do we decide which metadata fields to include?

• **What kind of information do we want to extract from the data?**	We need to know what is clinically or biologically relevant for our experimental hypothesis.
• **What is the experimental design?**	Do we have multiple batches, how many conditions, which treatment dose, etc.
• **Are there confounding elements?**	We need to document any condition or element that could alter or influence the results.
• **How many layer of information are there in our design?**	For complex studies, there might be the need to layer metadata and connect them through unique keys (e.g. patient's info, sample info, treatment info, etc.)

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How should a biobank record be structured?

Part 3 — Biobank Basics

A **biobank** is, at its core, a system for keeping three linked things in sync for every sample, indefinitely:

- **The physical sample** (tumor tissue, blood, DNA, RNA — stored, tracked, and eventually consumed by analyses like the one in this tutorial).
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- **Sample eligibility**: samples that lacks crucial information are no longer useful. In pediatrics, this could mean hindering years of research. In oncology or rare diseases, patients are few, samples even less, and gaining a small but usable cohort often takes years or the combined effort of multiple facilities. When age so dramatically impacts variability this process gets more difficult.

Part 4 — Hands-On: Build & Debug a Mini Biobank Record



EX. 2

The mini biobank exercise table — **2 records to complete, 6 records each hiding one error**.

patient_id	date_of_birth	age_years	sex	diagnosis	collection_date	sample_type	consent_status
PT-2031	03/01/2024						
PT-2032	22/11/2019						
PT-2032	30/06/2016	8	M	Ewing sarcoma	10/09/2024	Tumor tissue	Consented
PT-2033	14/05/2023	18	F	Retinoblastoma	02/11/2024	Tumor tissue	Consented
PT-2034	08/02/2020	5		Wilms tumor	01/03/2025	Tumor tissue	Consented
PT-2035	19/07/2022	2	M	Neuroblastoma	05/12/2021	Tumor tissue	Consented
PT-2036	27/09/2015	9	F	kid kidney tumor	14/10/2024	Tumor tissue	Consented
PT-2037	01/12/2018	6	M	Osteosarcoma	20/01/2025	Tumor tissue	

*Referral note — **record 1** (date of birth 3 January 2024). Female infant. Presented with a palpable abdominal mass; imaging and biopsy were consistent with neuroblastoma. A tumor tissue sample was biobanked on 20 March 2025, after the parents signed the institutional consent form.*

*Referral note — **record 2** (date of birth 22 November 2019). Male child. Diagnosed with osteosarcoma of the left femur; a tumor biopsy sample was biobanked on 5 June 2024, following parental consent.*

Part 4 — Hands-On: Build & Debug a Mini Biobank Record



EX. 3

Here are three experimental designs. Chose at least one and write down, based on what you consider relevant, which metadata fields would you include and how would you structure them.

Design 1: comparing treatment vs control using a certain drug, testing on three different cell lines and harvesting after 24, 36, 48 hours.

Design 2: comparing uninjured skin samples vs burn skin sample in pediatric patients to identify biomarkers and their impact in disease progression.

Design 3: a multicentric study recruits 200 subjects (100 controls, 100 patients). The goal is to study the association of a specific biomarker with a specific tumor. A biological sample is collected from each subject.

Part 4 — Hands-On: Build & Debug a Mini Biobank Record



EX. 3

Design 1: comparing treatment vs control using a certain drug, testing on three different cell lines and harvesting after 24, 36, 48 hours.

Sample_id	Cell_line	Group	Treatment	Dose	Timepoint	Replicate	Batch

Part 4 — Hands-On: Build & Debug a Mini Biobank Record



EX. 3

Design 2: comparing uninjured skin samples vs burn skin sample in pediatric patients to identify biomarkers and their impact in disease progression.

Sample_id	Subject_id	Age	Sex	Group	Burn_cause	Days_post_burn	Severity	TBSA%	Wound_location

Part 4 — Hands-On: Build & Debug a Mini Biobank Record



EX. 3

Design 3: a multicentric study recruits 200 subjects (100 controls, 100 patients). The goal is to study the association of a specific biomarker with a specific tumor. A biological sample is collected from each subject.

PATIENT								
Subject_id	Age	Sex	Group	Diagnosis	Stage	Recruitment_site	Treatment	Survival

SAMPLE						
Subject_id	Sample_id	Specimen_type	Collection_status	Collection_date	Assay_type	Biomarker